



Structural credit risk modelling with Hawkes jump diffusion processes



Yong Ma^{a,*}, Weidong Xu^b

^a College of Finance and Statistics, Hunan University, Changsha, 410079, PR China

^b School of Management, Zhejiang University, Hangzhou, 310085, PR China

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ABSTRACT

To describe the unexpectedness of default and especially default clustering in the framework of Merton's structural default, we propose a novel jump diffusion model for the firm's value. In this model, the jumps, which reflect the systematic risk common to all firms and an idiosyncratic risk, arrive dependently and they are described by self-exciting Hawkes processes rather than the classical Poisson processes. Some classical models are the special cases of the proposed model. The analytical solution to the value of the firm is derived. Numerical analysis shows that Hawkes jump diffusion model can better explain the behavior of default clustering than Poisson jump diffusion model.

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1. Introduction

Default clustering is repeatedly observed in financial market and has been empirically verified in the literature [1–4]. Some reduced-form credit risk models have recently been created to describe the behavior of default clustering [5–8]. Even so, it is still deserved to study default clustering using structural models, because structural models can be explained by economic fundamentals while reduced-form models cannot. To the best of our knowledge, however, structural models have seldom been constructed to serve this purpose. The challenge for structural models to do so lies in that they have to model the unexpectedness of default as well when they are used to fit default clustering, because people have extensively recognized that default time is unpredictable and any reduced-form model is capable of producing unexpected default time. In this paper we will construct a novel structural default model for characterizing unexpected default and importantly it is more appropriate to explain the phenomenon of default clustering than the classic structural models.

The structural approach to credit risk was initiated by Merton [9], and Black and Cox [10], and was subsequently extended in various directions. As shown in [11], a major issue within structural models is the modeling of the dynamics of the firm's value. A primary assumption underlying the classical structural models is that the firm's value is characterized by geometric Brownian motion. Zhou [12] analyzed the default correlation based on Black–Cox model. In this circumstance, due to the continuity of the firm value, default time is predictable and thereby firms will not default unexpectedly. However, it cannot explain some facts in reality such as the sudden bankruptcy of Enron in 2001 which was once named by Fortune America's Most Innovative Company for six consecutive years, and large credit spreads for some corporate bonds with short maturities. To comply with the reality, people may either adopt reduced-form default model or incorporate jumps into the dynamics of the firm value process to produce unpredictable default time. Because reduced-form approach is not adopted in this paper, we will not go into depth on it. Among the structural models on the basis of jump diffusion processes, there are Zhou [13],

* Corresponding author. Tel.: +86 13549658676.

E-mail addresses: yyma@hnu.edu.cn (Y. Ma), weidxu@zju.edu.cn (W. Xu).

Baxter [14], Willemann [15], Hurd [16] and so on. In these models, Poisson processes or more generalized Lévy processes are used to characterize the jumps. As is well known, Lévy processes have the property of independence of increments; that is, jumps are uncorrelated. But Aït-Sahalia et al. [4] argued that what makes financial crisis take place is typically not the initial jump, but the amplification that occurs subsequently over hours or days, and the fact that other markets become affected as well. Thus Poisson jump diffusion and even Lévy models fail to replicate jump clustering both in time and across markets. As a result, how to build a model of the firm's value which can describe jump clustering becomes one of our concerns.

On the other hand, the collapse of some major financial institutions in the 2007–2008 subprime mortgage crisis such as Lehman Brothers, Fannie Mae, Freddie Mac, Merrill Lynch and Bear Sterns shows the emergence of default clustering in the same sector. Besides, the fact that many European countries have experienced sovereign debt crisis in recent years exemplifies default clustering in the same region. As pointed out in [7], default clustering mainly results from two aspects. One aspect is that the firms are exposed to a common risk factor which impacts all entities in the portfolio. The other is the contagion effect which makes a default directly impair the health of other firms. We do not aim to explore the sources of default clustering, on which more detailed study can be found in [17] and references therein. Because the existence of default clustering is now beyond dispute, a natural academic question arises: How can we develop better models to incorporate the default clustering effect when studying portfolio credit risk?

In this paper, we build a Hawkes jump diffusion model for the firm's value where jumps are characterized by Hawkes processes. Since for Hawkes processes the occurrence of a jump will accelerate the arrival of future jumps in probability, they are capable of reproducing jump clustering. Thus the proposed model has advantage over classical jump diffusion structural models. In addition, we assume that jumps for every firm's value are described by an individual Hawkes process and a common Hawkes process. The individual Hawkes processes are used to capture the firm-specific jump clusters resulting from the idiosyncratic shocks such as continuous bad news, frequent investment losses. Analogously, the common Hawkes process is used to capture the jump clusters growing out of the systematic shocks such as financial crisis. This means that defaults are implicitly linked by the correlated Brownian motion and the common Hawkes process. We derive the closed-form expression for the firm value and the expression for default correlation. Numerical examples show that default correlations in the proposed model are larger than in Poisson jump diffusion process model, which implies that our model is more suitable to fit default clustering effect. In a word, our model is not only able to reflect jump clustering in firm value, but it is also a better candidate than standard jump diffusion models for fitting default clustering.

The remainder of the paper is organized as follows. Section 2 builds the firm's value model, and gets the analytic solution to it and the expression for default correlation. Section 3 compares the proposed model with two classical models with respect to the probability of default and default correlation; sensitivity analysis is also made in this section. Section 4 provides a brief conclusion.

2. The model

In this section, self-exciting Hawkes processes are used as the jump components to construct a jump diffusion model for the value of the firm.

2.1. Default model set-up

We only take the case of two entities into consideration, because it can be easily extended to the general case. Define three independent Hawkes processes as follows:

$$N_t = \sum_{i=1}^{\infty} 1(T_i \leq t), \quad N_{n,t} = \sum_{i=1}^{\infty} 1(T_{n,i} \leq t), \quad n = 1, 2 \quad (1)$$

where $\{T_i\} > 0$ and $\{T_{n,i}\} > 0$ are the jump times of N_t and $N_{n,t}$, respectively. The intensity processes of Hawkes processes N_t , $N_{n,t}$ are, respectively,

$$\lambda_t = \lambda_0 + \theta \int_0^t e^{-\delta(t-s)} dN_s = \lambda_0 + \theta \sum_{T_i < t} e^{-\delta(t-T_i)}, \quad (2)$$

$$\lambda_{n,t} = \lambda_{n,0} + \theta_n \int_0^t e^{-\delta_n(t-s)} dN_{n,s} = \lambda_{n,0} + \theta_n \sum_{T_{n,i} < t} e^{-\delta_n(t-T_{n,i})}, \quad (3)$$

where $\lambda_0, \theta, \delta, \lambda_{n,0}, \theta_n, \delta_n$ are positive constants.¹ In addition, for model comparison, suppose $0 < \theta < \delta$ and $0 < \theta_n < \delta_n$, then it follows from [18,19] that the expected numbers of jumps between $[0, t]$ are $\mathbb{E}[N_t] = \frac{\lambda_0 t}{1-\theta/\delta}$, $\mathbb{E}[N_{n,t}] = \frac{\lambda_{n,0} t}{1-\theta_n/\delta_n}$, $n = 1, 2$.

¹ We use the Hawkes process with exponential decay rather than other feasible models (such as the epidemic-type aftershock sequence model) because it is a popular specification in literature and because, as noted in [20], it makes the Hawkes process become a Markov process, thus contributing to computation efficiency.

In this paper, N_t is used as the common Hawkes process, and it reflects the jumps growing out of systematic events such as financial crisis, which affects all the entities in financial system; $N_{n,t}$ are used to represent the jumps related to firm-specific events, such as the obtaining of a new patent and the release of an annual report. The intensity process λ_t of Hawkes process N_t means that if a systematic shock, which leads to a jump, occurs in financial market, the intensity will increase immediately by θ and then the increment decays with exponential rate δ . In other words, the occurrence of a systematic shock will excite the occurrence of future systematic shocks, and so the jumps will cluster to some extent. That is why self-exciting Hawkes processes might be suitable to characterize the phenomenon of jump clustering. Besides, the larger θ is or (and) the smaller δ is, the more clustering the jumps will be. In contrast, the jumps in Poisson processes will not affect each other because of the independence of increments. However, when θ goes to zero, the excitedness of Hawkes process N_t will get weaker and weaker, and the process finally leads to a Poisson process with constant intensity λ .

Let $V_{1,t}$, $V_{2,t}$ be the firm values and they are driven by Hawkes jump diffusion processes where jumps are described by two independent Hawkes processes; one is used to capture systematic shocks, and the other firm-specific or idiosyncratic shocks. Specifically, for $n = 1, 2$,

$$\begin{aligned} dV_{n,t}/V_{n,t-} &= \mu_n dt + \sigma_n dW_{n,t} + d\left[\sum_{i=1}^{N_t+N_{n,t}} Z_{n,i} - \beta_n \int_0^t \lambda_s ds - \beta_n \int_0^t \lambda_{n,s} ds\right] \\ &= \left[\mu_n - \beta_n(\lambda_0 + \lambda_{n,0} + \theta \sum_{T_i < t} e^{-\delta(t-T_i)} + \theta_n \sum_{T_{n,i} < t} e^{-\delta_n(t-T_{n,i})})\right] dt + \sigma_n dW_{n,t} + d\sum_{i=1}^{N_t+N_{n,t}} Z_{n,i} \end{aligned}$$

where $\{Z_{n,i}\}$ are the magnitudes of the jumps and they are identically and independently distributed, $\beta_n = \mathbb{E}[Z_{n,i}]$, and $W_{1,t}$, $W_{2,t}$ are two correlated Brownian motions with correlation coefficient ρ . As a result, the two firm values are linked by the common Hawkes process and correlated Brownian motions.

2.2. The solutions to firm values and default correlation

In the following we derive the analytical solution to the firm values and the expression of default correlation.

Proposition 2.1.

$$\begin{aligned} X_{n,t} &= V_{n,0} \exp\left\{\sigma_n W_{n,t} + \left(\mu_n - \beta_n \lambda_0 - \beta_n \lambda_{n,0} - \frac{1}{2}\sigma_n^2\right)t\right. \\ &\quad \left.+ \sum_{i=1}^{N_t} \frac{\beta_n \theta}{\delta} [e^{-\delta(t-T_i)} - 1] + \sum_{i=1}^{N_{n,t}} \frac{\beta_n \theta_n}{\delta_n} [e^{-\delta_n(t-T_{n,i})} - 1]\right\}, \end{aligned}$$

and

$$J_{n,t} = \prod_{i=1}^{N_t+N_{n,t}} (1 + Z_{n,i}), \quad (4)$$

where the product over an empty set is taken to be one. Then

$$V_{n,t} = X_{n,t} J_{n,t}. \quad (5)$$

Proof. It is clear that, for $n = 1, 2$, $X_{n,t}$ are continuous stochastic processes. Then, according to Itô's formula, we have

$$dX_{n,t} = \left[\mu_n - \beta_n(\lambda_0 + \lambda_{n,0} + \theta \sum_{i=1}^{N_t} e^{-\delta(t-T_i)} + \theta_n \sum_{i=1}^{N_{n,t}} e^{-\delta_n(t-T_{n,i})})\right] X_{n,t} dt + \sigma_n X_{n,t} dW_{n,t}. \quad (6)$$

Besides,

$$dJ_{n,t} = J_{n,t-} \cdot d\left(\sum_{i=1}^{N_t+N_{n,t}} Z_{n,i}\right). \quad (7)$$

Since X_n is continuous and J_n is a pure jump process, according to Itô's formula for semimartingales,

$$\begin{aligned} dV_{n,t} &= d(X_{n,t} J_{n,t}) = X_{n,t-} dJ_{n,t} + J_{n,t-} dX_{n,t} + d[X_n, J_n](t) \\ &= X_{n,t-} dJ_{n,t} + J_n(t-) dX_{n,t}. \end{aligned} \quad (8)$$

Substituting (6) and (7) into (8) yields the desired result. $\square$

Note that due to the fact that the sum of two independent Poisson processes is still a Poisson process, if $\theta = \theta_n = 0$, then the firm value V_n follows a Poisson jump diffusion process with intensity $\lambda_0 + \lambda_{n,0}$.

In the framework of Merton's structural default, the default of a firm is triggered at the expiration date of a debt contract when the firm's value falls to the threshold level which is mainly determined by the firm's debt. Following Black and Cox [10], for each firm n we assume that the threshold level $C_{n,t}$ takes an exponential model: $C_{n,t} = e^{\kappa_n t} D_i$. The default boundary $C_{n,t}$ can be interpreted as the minimum firm value required by the safety covenant of a debt contract, and it takes the exponential form since the expected debt value usually has this form and κ_n is regarded as the growth rate of firm n 's liabilities.

Proposition 2.2. Let T be the expiration date of a debt contract, and $C_{1,t}$, $C_{2,t}$ the default boundaries of the two firms, respectively. Assume that the jump sizes of each firm's value follow a log-normal distribution²; specifically, $\log(1 + Z_{n,i}) \sim N(a_n, b_n)$, $i = 1, 2, \dots$, and let $\beta_n = \exp(a_n + \frac{1}{2}b_n^2) - 1$. Then the probabilities of default, under Merton's structural approach, are

$$p_n(T) = \mathbb{E}[\Phi(f_n(R_n, K_n))], \quad n = 1, 2, \quad (9)$$

where $\Phi(\cdot)$ is the cumulative distribution function (CDF) of the standard normal distribution,

$$K_n = N_T + N_{n,T}, \quad (10)$$

$$R_n = \prod_{i=1}^{N_T} \exp\left\{\frac{\beta_n \theta}{\delta} [e^{-\delta(T-T_i)} - 1]\right\} \prod_{i=1}^{N_{n,T}} \exp\left\{\frac{\beta_n \theta_n}{\delta_n} [e^{-\delta_n(T-T_{n,i})} - 1]\right\}, \quad (11)$$

and

$$f_n(R_n, K_n) = -\frac{\log(R_n V_{n,0}/D_n) + (\mu_n - \kappa_n - \beta_n \lambda_0 - \beta_n \lambda_{n,0} - \frac{1}{2}\sigma_n^2)T + a_n K_n}{\sqrt{\sigma_n^2 T + b_n^2 K_n}}. \quad (12)$$

Let $\bar{\rho} := \bar{\rho}(K_1, K_2) = \frac{\rho \sigma_1 \sigma_2 T}{\sqrt{(\sigma_1^2 T + b_1^2 K_1)(\sigma_2^2 T + b_2^2 K_2)}}$. Then the default correlation is

$$\rho_{12}(T) = \frac{\mathbb{E}[\Phi_2(f_1(R_1, K_1), f_2(R_2, K_2); \bar{\rho})] - \mathbb{E}[\Phi(f_1(R_1, K_1))]\mathbb{E}[\Phi(f_2(R_2, K_2))]}{\sqrt{p_1(T)(1-p_1(T))}\sqrt{p_2(T)(1-p_2(T))}}, \quad (13)$$

where $\Phi_2(\cdot, \cdot; \bar{\rho})$ is the CDF of the bivariate norm-distributed random variable with mean zero and covariance matrix $\begin{pmatrix} 1 & \bar{\rho} \\ \bar{\rho} & 1 \end{pmatrix}$.

Proof. According to Proposition 2.1, for $n = 1, 2$, we have

$$\log\left(\frac{V_{n,T}}{R_n V_{n,0}}\right) = \sigma_n W_{n,T} + \left(\mu_n - \beta_n \lambda_0 - \beta_n \lambda_{n,0} - \frac{1}{2}\sigma_n^2\right)T + \sum_{i=1}^{K_n} \log(1 + Z_{n,i}).$$

Let $\mathcal{F}_{n,t} = \sigma\{N_s \cup N_{n,s} : 0 \leq s \leq t\}$, $n = 1, 2$, $\mathcal{F}_t = \mathcal{F}_{1,t} \cup \mathcal{F}_{2,t}$. Because $W_{n,T}$ and $Z_{n,i}$ are independent, then given information $\mathcal{F}_{n,T}$,

$$\sigma_n W_{n,T} + \sum_{i=1}^{K_n} \log(1 + Z_{n,i}) := X_n \sim N(a_n K_n, \sigma_n^2 T + b_n^2 K_n).$$

Since $W_{n,T} \sim N(0, T)$, $dW_{1,t} dW_{2,t} = \rho dt$, and $\{Z_{1,i}\}$ and $\{Z_{2,i}\}$ are independent, given $\mathcal{F}_T$,

$$\rho(X_1, X_2) := \bar{\rho} = \frac{\rho \sigma_1 \sigma_2 T}{\sqrt{\sigma_1^2 T + b_1^2 K_1} \sqrt{\sigma_2^2 T + b_2^2 K_2}}.$$

Given information $\mathcal{F}_{n,T}$, then the conditional probability of default is

$$\begin{aligned} \mathbb{P}\{V_{n,T} \leq C_{n,T} | \mathcal{F}_{n,T}\} &= \mathbb{P}\left\{\log\left(\frac{V_{n,T}}{R_n V_{n,0}}\right) \leq \log\left(\frac{C_{n,T}}{R_n V_{n,0}}\right) | \mathcal{F}_{n,T}\right\} \\ &= \mathbb{P}\left\{X_n \leq \log\left(\frac{D_n}{R_n V_{n,0}}\right) - \left(\mu_n - \kappa_n - \beta_n \lambda_0 - \beta_n \lambda_{n,0} - \frac{1}{2}\sigma_n^2\right)T | \mathcal{F}_{n,T}\right\} \\ &= \Phi(f_n(R_n, K_n)). \end{aligned}$$

As a result, the unconditional probability of default of firm n is

$$p_n(T) = \mathbb{E}[\Phi(f_n(R_n, K_n))]. \quad (14)$$

² First of all, it is very common in literature to assume that jump sizes follow log-norm distribution; see, e.g., [21,22,13,23,24] and literature therein. Secondly, this assumption can make the solution to default correlation most tractable. The less popular distribution for jump size is double exponential distribution.

Now we show the second part of this theorem. Given $\mathcal{F}_T$, the conditional probability of joint default is

$$\begin{aligned}\mathbb{P}\{V_{1,T} \leq C_{1,T}, V_{2,T} \leq C_{2,T} | \mathcal{F}_T\} &= \mathbb{P}\left\{\log \frac{V_{n,T}}{R_n V_{n,0}} \leq \log \frac{C_{n,T}}{R_n V_{n,0}}, n = 1, 2 | \mathcal{F}_T\right\} \\ &= \mathbb{P}\left\{X_n \leq \log \frac{D_n}{R_n V_{n,0}} - \left(\mu_n - \kappa_n - \beta_n \lambda_0 - \beta_n \lambda_{n,0} - \frac{1}{2} \sigma_n^2\right) T, n = 1, 2 | \mathcal{F}_T\right\} \\ &= \Phi_2(f_1(R_1, K_1), f_2(R_2, K_2); \bar{\rho}).\end{aligned}\quad (15)$$

Thus the unconditional probability of joint default is

$$p_{12}(T) = \mathbb{P}\{V_{1,T} \leq D_1, V_{2,T} \leq D_2\} = \mathbb{E}[\Phi_2(f_1(R_1, K_1), f_2(R_2, K_2); \bar{\rho})].$$

Because default correlation

$$\rho_{12}(T) = \frac{p_{12}(T) - p_1(T)p_2(T)}{\sqrt{p_1(T)(1-p_1(T))}\sqrt{p_2(T)(1-p_2(T))}},\quad (16)$$

substituting Eqs. (14) and (15) into it leads to the desired result immediately. $\square$

When $\lambda = \lambda_n = 0$, $\theta = \theta_n = 0$, the dynamics of firm value reduces to geometric Brownian motion, and the probability of default becomes

$$p_n(T) = \Phi\left(-\frac{\log(V_{n,0}/D_n) + (\mu_n - \kappa_n - \sigma_n^2/2)T}{\sqrt{\sigma_n^2 T}}\right),\quad (17)$$

which agrees with the corresponding result by letting $\lambda = \lambda_n = 0$, $\theta = \theta_n = 0$ in Eqs. (10), (11) and (17), i.e. $\Phi(f_n(1, 0))$. In this case, the values of the two firms are linked by correlated Brownian motions, and the probability of joint default in $[0, T]$ is

$$\begin{aligned}p_{12}(T) &= \Phi_2(f_1(1, 0), f_2(1, 0); \rho) \\ &= \Phi_2\left(-\frac{\log(V_{1,0}/D_1) + (\mu_1 - \kappa_1 - \sigma_1^2/2)T}{\sqrt{\sigma_1^2 T}}, -\frac{\log(V_{2,0}/D_2) + (\mu_2 - \kappa_2 - \sigma_2^2/2)T}{\sqrt{\sigma_2^2 T}}; \rho\right).\end{aligned}\quad (18)$$

Thus, for the classical diffusion model, substituting (17) and (18) into (16) gives rise to the default correlation

$$\rho_{12}(T) = \frac{\Phi_2(f_1(1, 0), f_2(1, 0); \rho) - \Phi(f_1(1, 0))\Phi(f_2(1, 0))}{\sqrt{\Phi(f_1(1, 0))(1 - \Phi(f_1(1, 0)))}\sqrt{\Phi(f_2(1, 0))(1 - \Phi(f_2(1, 0)))}}.$$

In addition, when $\theta = \theta_n = 0$, the dynamics of firm value degenerates to jump diffusion process where jumps are described by Poisson processes, and the probability of default is

$$p_n(T) = e^{-(\lambda_0 + \lambda_{n,0})T} \sum_{k=0}^{\infty} \frac{((\lambda_0 + \lambda_{n,0})T)^k}{k!} \cdot \Phi\left(-\frac{\log(V_{n,0}/D_n) + (\mu_n - \kappa_n - \beta_n \lambda_0 - \beta_n \lambda_{n,0} - \frac{1}{2} \sigma_n^2)T + a_n k}{\sqrt{\sigma_n^2 T + b_n^2 k}}\right).\quad (19)$$

In this case, the values of the two firms are linked by correlated Brownian motion and a common Poisson process, and the probability of joint default becomes

$$p_{12}(T) = e^{-(\lambda_0 + \lambda_{1,0} + \lambda_{2,0})T} \sum_{k=0}^{\infty} \sum_{k_1=k}^{\infty} \sum_{k_2=k}^{\infty} \frac{(\lambda_0 T)^k (\lambda_{1,0} T)^{k_1-k} (\lambda_{2,0} T)^{k_2-k}}{k!(k_1-k)!(k_2-k)!} \cdot \Phi_2(f_1(1, k_1), f_2(1, k_2); \bar{\rho}).\quad (20)$$

Substituting (19) and (20) into (16), we then have the default correlation for Poisson jump diffusion model. Therefore, the proposed model includes geometric Brownian model and Poisson jump diffusion model as the special cases.

In order to completely determine the default probability and the default correlation in Proposition 2.2, to the best of our knowledge, Monte Carlo method is the sole method available in the literature now. An important step when using Monte Carlo method to derive the default correlation is to simulate the paths of Hawkes processes. Once the paths of Hawkes processes (namely the occurrence times of jumps) are simulated, the default probability and the default correlation can be approximated based on the sample means of $\Phi(f_n(R_n, K_n))$ and $\Phi_2(f_1(R_1, K_1), f_2(R_2, K_2); \bar{\rho})$.

There are already several algorithms proposed to simulate the paths of Hawkes processes (see [25–27]). However, as shown in [27], the current standard algorithm for simulating Hawkes processes is still the conventional thinning method. Therefore, we also apply thinning algorithm proposed by Ogata [28] to simulate the paths of Hawkes processes. Thinning algorithm for N_t is shown below, and those for $N_{n,t}$, $n = 1, 2$, are the same.

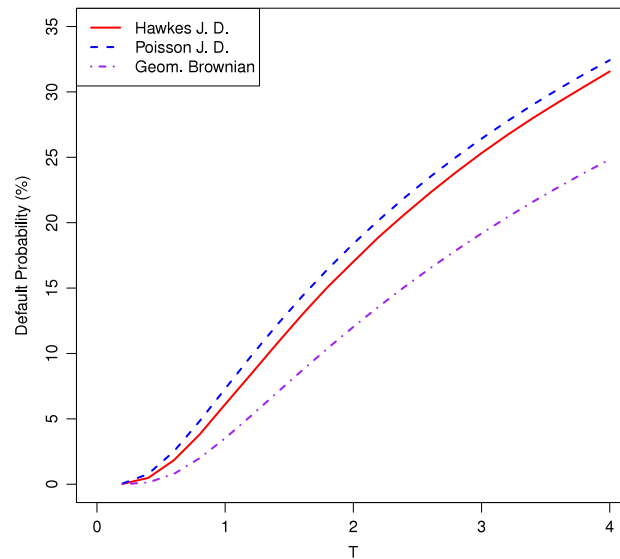


Fig. 1. The probability of default against risk horizon under the models of geometric Brownian, Poisson jump diffusion and Hawkes jump diffusion. $a_1 = a_2 = 0$, $b_1 = b_2 = 0.1$.

Thinning Algorithm

1. Set $\lambda \leftarrow \lambda_0$, $n \leftarrow 1$.
2. Generate a standard uniformly distributed variate U , i.e., $U \sim U(0, 1)$. Let $s \leftarrow -\ln U/\lambda$. If $s \leq T$, then $t_1 \leftarrow s$; else go to the last step.
3. Set $n \leftarrow n + 1$, $\lambda \leftarrow \lambda_{t_{n-1}} + \theta$. If $s \leq T$, then go to the next step; else go to the last step.
4. Generate $U \sim U(0, 1)$, and let $s \leftarrow s - \ln U/\lambda$.
5. Generate $V \sim U(0, 1)$. If $V \leq \lambda_s/\lambda$, a new event occurs at time s , i.e., $t_n \leftarrow s$, and then return to step 3; else return to step 4.
6. Stop simulation. The occurrence times of the events are $t_1, t_2, \dots, t_n$.

Note that although the traditional thinning algorithm is sufficient to derive the default correlation, more contemporary numerical techniques (see, e.g., [29–34]) should be adopted for reducing the computational burden in the future research when studying the default correlation under the framework of the first-passage-time structural model.

3. Numerical analysis

In this section, model comparison and sensitivity analysis are mainly made to explore the effect of jump clustering on the probability of default and default correlation. In the base case, we assume that $V_{1,0}/D_1 = V_{2,0}/D_2 = 1.8$. They can be seen as the proxies of the credit quality of two firms, respectively, and their values are the same as those in [12]. For model comparison, let $\sigma_1 = \sigma_2 = 0.3$, $\rho = 0.4$, which also coincide with the values in [12]. Besides, let $\mu_1 = \kappa_1$, $\mu_2 = \kappa_2$. Economically, $\mu_n = \kappa_n$ indicates that the value of the firm debt and the firm value have the same growth rate on average. Thereby, this means that the firm stays in the stage of stable development and it bears a constant leverage ratio. Finally, according to the empirical research in [4], jumps in US financial market occur four times each year on average, we let $\lambda_0 = \lambda_{1,0} = \lambda_{2,0} = 1$, $\theta = \theta_1 = \theta_2 = 1$, $\delta = \delta_1 = \delta_2 = 2$, which means that half of jumps are from systematic factors and half from idiosyncratic factors. To explore the effect of jump clustering, we will compare the proposed model with Poisson jump diffusion model and the classical Merton structural model. For consistency of the expected number of jumps in unit time, for the common jumps and firm-specific jumps the intensities of Poisson processes are taken to be $\lambda_0/(1 - \theta/\delta)$ and $\lambda_{n,0}/(1 - \theta_n/\delta_n)$, respectively. Note that in the following model comparison and sensitivity analysis, we only change the value of one of the parameters each time and keep the values of the other parameters unchanged as given in the base case.

3.1. Analysis of probability of default

The comparison of the probabilities of default under the proposed model and two reference models, which are Hawkes jump diffusion (Hawkes J. D.), geometric Brownian (Geom. Brownian) and Poisson jump diffusion (Poisson J. D.), is shown in Fig. 1. It follows from this figure that jump risk increases the probability of default, but jump clustering decreases it. Besides, the gaps of the probabilities of default between these models become more and more significant as risk horizon increases.

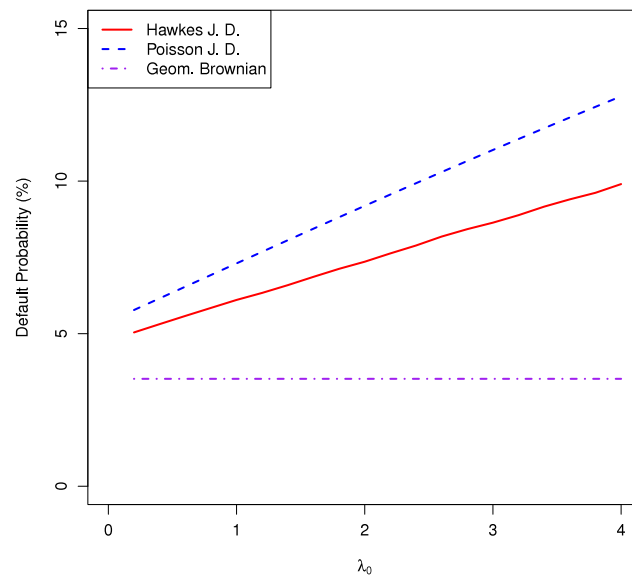


Fig. 2. The probability of default against the initial intensity of the common Hawkes process and Poisson process. The unspecified parameter value used here is $T = 1$.

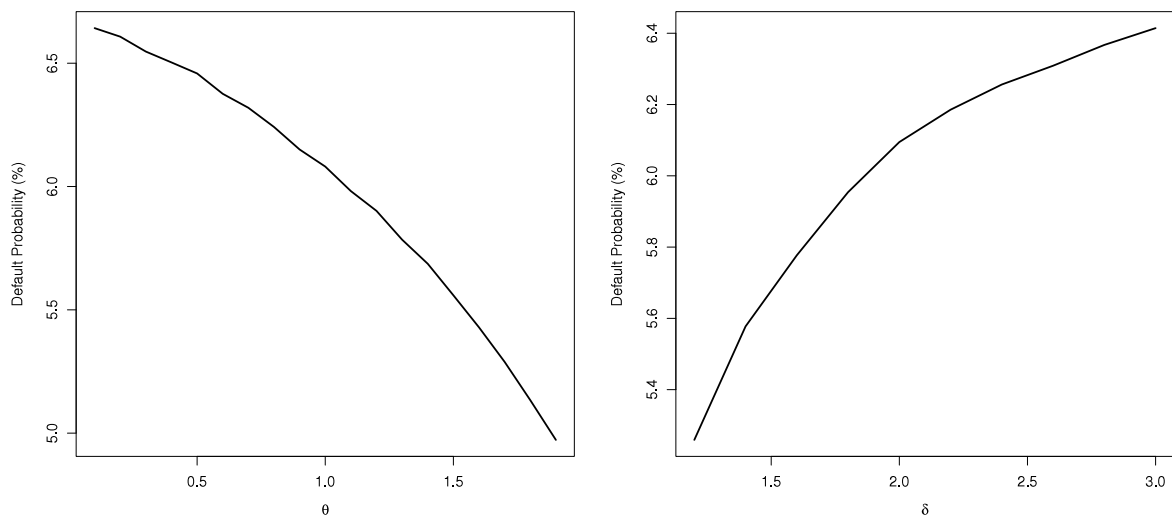


Fig. 3. The effects of the parameters θ and δ which characterize jump clustering risk on the probability of default with two jumps on average in unit time. Left: $\lambda_0 = 2 - \theta$, right: $\lambda_0 = 2(1 - 1/\delta)$.

As presented in Fig. 2, the probability of default is an increasing function of the initial intensity, under the models which include jump components. This result also leads to the conclusion that the likelihood of default tends to get larger as jump risk increases. Fig. 3 illustrates the sensitivity of the probability of default to the parameters of the common Hawkes process which describes jump clustering risk. Because the larger parameter θ (δ) is, the higher (lower) jump clustering risk is, we conclude that jump clustering will lower the occurrence of default, that is, the higher jump clustering risk is, the less the probability of default is. Therefore, when calibrated to the market-implied probability of default (e.g., the probability implied in the price of the defaultable bond or credit default swap (CDS)), if the diffusion part is fixed, our model has more jumps than Poisson jump diffusion model on average.

3.2. Analysis of default correlation

This subsection serves to compare models and analyze the sensitivity of default correlation with respect to some parameters. From Fig. 4, we know that default correlation increases as risk horizon increases, which is obvious by the definition of default correlation. Additionally, geometric Brownian model has the largest default correlation among the three models, Hawkes jump diffusion model follows and Poisson jump diffusion model has the least. It follows that, contrary to the

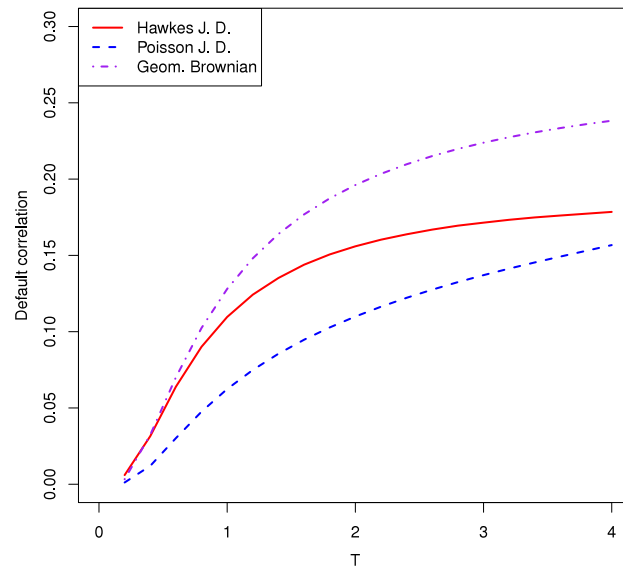


Fig. 4. Default correlation against risk horizon under the models of geometric Brownian, Poisson jump diffusion and Hawkes jump diffusion.

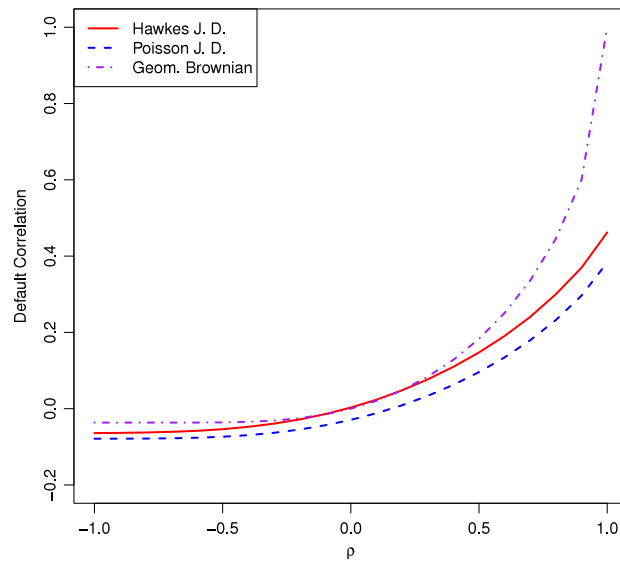


Fig. 5. Default correlation against correlation coefficient under the three models.

effects of jump risk and jump clustering on probability of default, jump risk lowers default correlation and jump clustering improves it.

Fig. 5 exhibits the effect of correlation coefficient of Brownian motions on default correlation. In terms of default correlation, we find that the proposed model and two reference models do not make considerable difference as ρ varies from -1 to 0 . However, the difference between them becomes more and more significant as ρ increases from 0 to 1 . In particular, when ρ is larger than 0.5 , the gap between default correlations derived from diffusion model and jump diffusion models widens rapidly. What is more, for any ρ default correlation in Hawkes jump diffusion model is larger than that in Poisson jump diffusion model, while the relationship between default correlations in geometric Brownian model and Hawkes jump diffusion model is uncertain. Note that default correlation is increasing in the correlation of two Brownian motions and it is negative when ρ is negative, but it only changes slightly as ρ goes to -1 , which is because the total values of two firms are associated positively as well. This characteristic of default correlation for our model is not a shortcoming, because people are more concerned to the positively correlated defaults in portfolio risk management.

In Fig. 6, as λ_0 or $\lambda_0^{(1)}$, the initial intensity of Hawkes process, changes from 0 to 3 , the intensities of Poisson processes are adjusted correspondingly in order to ensure that the expected number of jumps is consistent. Fig. 6 indicates that default correlation in Hawkes jump diffusion process is always larger than that in Poisson jump diffusion process, and the difference

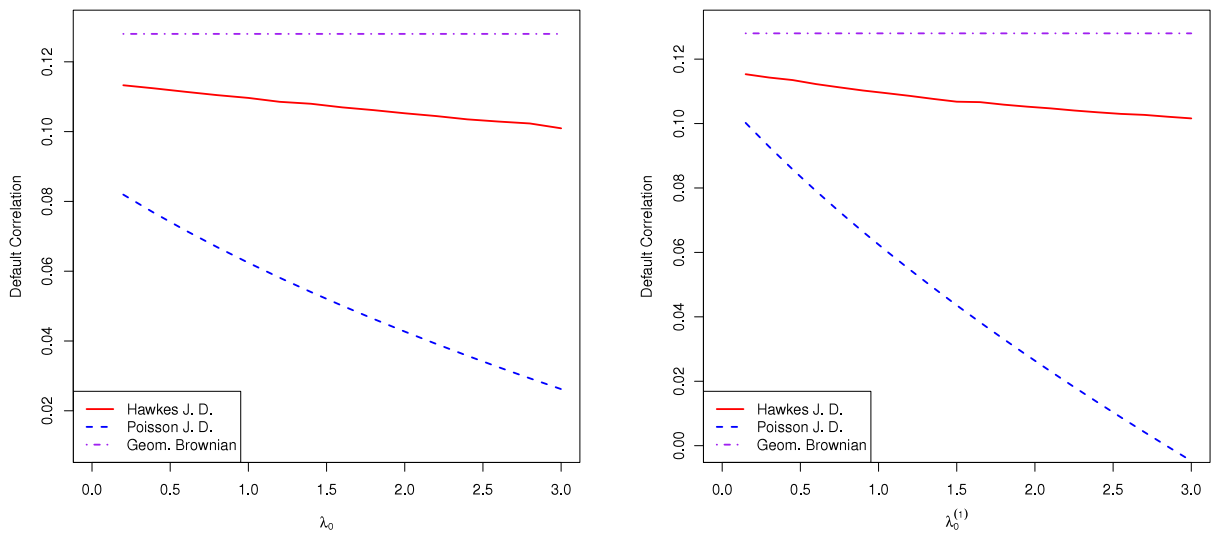


Fig. 6. Default correlation against the initial intensity of the common Hawkes process and Poisson process (left), and the case for the idiosyncratic processes (right).

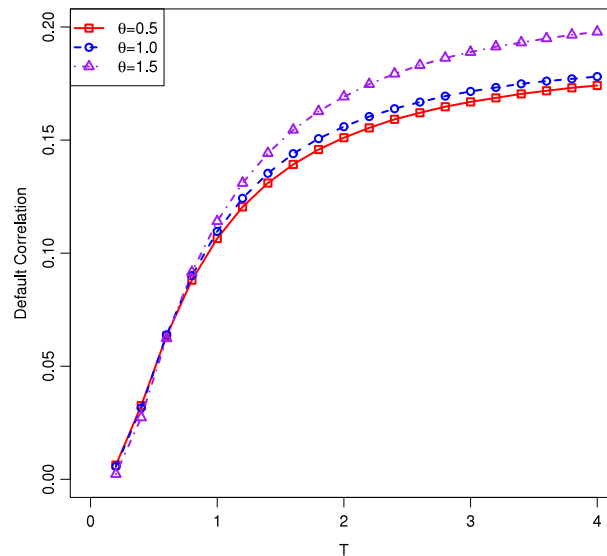


Fig. 7. The relation between default correlation and the magnitude of jump clustering of the common Hawkes process which has the same expected number of jumps as the base case.

gets more and more pronounced. This result implies that when the expected number of jumps increases, the positive impact of jump clustering on the correlation of default becomes more and more powerful. In other words, jump clustering increases default correlation and the influence becomes stronger as the emergence of jumps becomes more frequent. Fig. 6 explores the effect of jump clustering on default correlation by model comparison, while Figs. 7 and 8 explore the impact directly by the variation of parameters which characterize jump clustering. As θ or θ_1 increases, i.e., the magnitude of jump clustering becomes larger, the correlation between defaults becomes stronger. Furthermore, the discrepancy of default correlation for different magnitudes of jump clustering becomes more and more obvious as risk horizon becomes longer. It can be explained by the fact that the expected number of jumps is increasing in risk horizon, and as concluded from Fig. 6, the effect of jump clustering gets stronger as the expected number of jumps increases. On the other hand, as δ or δ_1 increases, i.e., jump clustering lowers, default correlation becomes weaker. It conforms to the fact that the impacts of $\theta(\theta_1)$ and $\delta(\delta_1)$ on jump clustering are opposite.³ Thereby we come to the conclusion that jump clustering, whatever it is systematic or

³ Due to the limitedness of the space, we do not show the figures about the effects of δ and δ_1 on default correlation. Besides, because of symmetry, the results for θ_1 and δ_1 also hold for θ_2 and δ_2 , respectively.

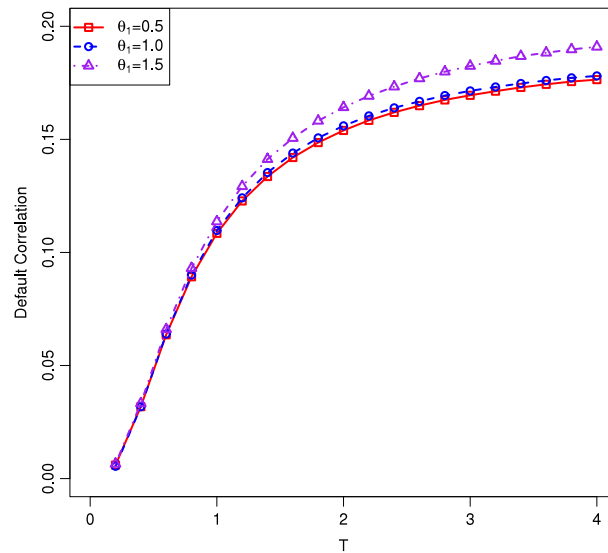


Fig. 8. The relation between default correlation and the magnitude of jump clustering of the idiosyncratic Hawkes process which has the same expected number of jumps as the base case.

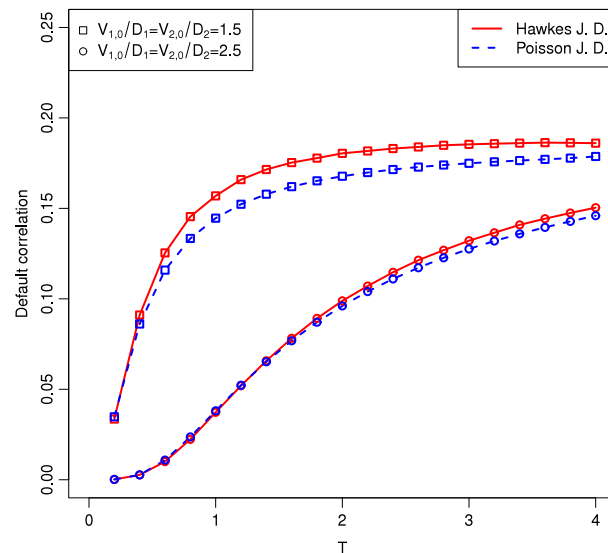


Fig. 9. The relation between default correlation and risk horizon for different credit qualities.

idiosyncratic, strengthens the dependence of default. In a word, for Hawkes jump diffusion model, the larger the expected number of jumps is, the larger the default correlation is.

As shown in the previous subsection, when we calibrate the two jump diffusion models to the implied default probabilities of the prices of credit assets, Hawkes jump diffusion model has more expected jumps than Poisson jump diffusion model. Furthermore, according to the conclusions in this subsection, we conclude that Hawkes jump diffusion model has larger default correlation than Poisson jump diffusion model with the same individual default probabilities and diffusion part. As a consequence, Hawkes jump diffusion model is a better candidate for describing default clustering. To test the robustness of the conclusion, comparison is also made for the films with two other credit qualities, $V_{1,0}/D_1 = V_{2,0}/D_2 = 1.8$ and $V_{1,0}/D_1 = V_{2,0}/D_2 = 2.5$. As shown in Fig. 9, for either model default correlation is always decreasing in the films' credit quality. In addition, no matter whether the credit quality is better or worse than the base case, the proposed model yields larger default correlation than Poisson jump diffusion model. This suggests that the conclusion is robust with respect to the credit quality of the films. Robustness with respect to the parameters related to the amplitude of jumps is also validated; see Fig. 10.

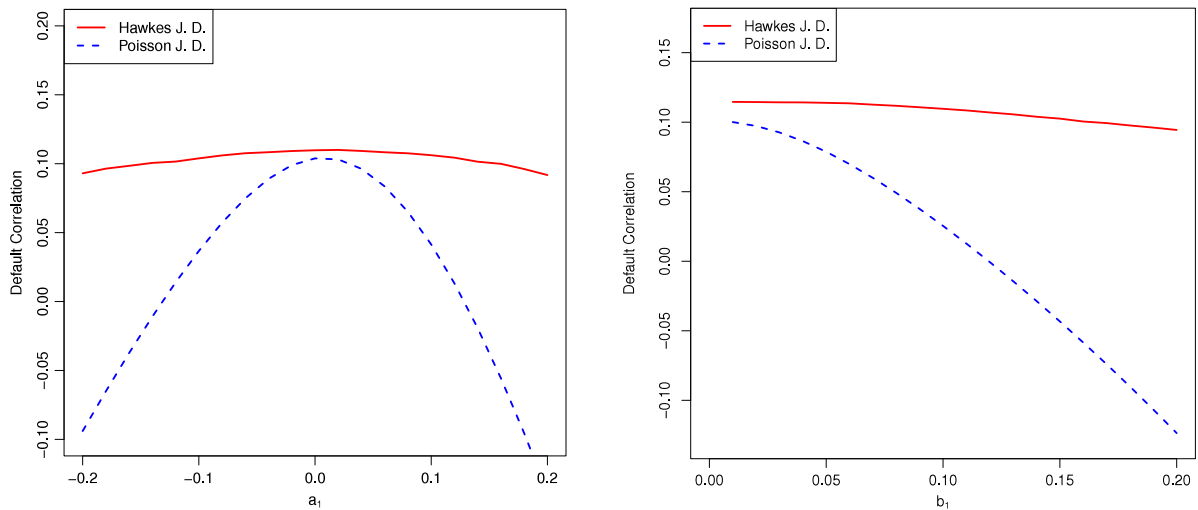


Fig. 10. The relation between default correlation and the parameters related to the amplitude of jumps. $T = 1$. Note that because of symmetry the results are the same for the parameters a_2 and b_2 , respectively.

4. Conclusion

Within the framework of Merton's structural default approach, we construct jump diffusion processes for the values of the assets, where jumps are described by self-exciting Hawkes processes rather than Poisson process. The dependence of default is built by a common Hawkes process and the correlated Brownian motions. By numerical analysis, we find that, firstly, jump risk increases the probability of default which has been shown repeatedly in literature, while jump clustering brings down the probability of default for any individual entity. Secondly, on contrary to the effects of jump risk and jump clustering on probability of default, the effect of jump risk on default correlation is negative and jump clustering contributes to the improvement of the correlation of default. Lastly, in terms of fitting default clustering, Hawkes jump diffusion model outperforms Poisson jump diffusion model.

Though we have shown that our model is appropriate to produce default clustering by numerical analysis, we have not formulated how to do model calibration and parameter estimation. Therefore, further justification by empirical analysis is still needed in the future study. Besides, one of the future research directions is to consider the Hawkes jump diffusion model within the framework of first-passage-time structural model and explore whether it is also a good candidate for modeling default clustering.

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